

Summary

Progressive Web App (PWA) Introduction

During this period, we made key implementation decisions regarding the future direction of the Eco Clean platform's mobile experience. As the web admin system became completer and more stable, we evaluated whether maintaining a separate mobile application was the most efficient and cost effective long-term approach. Based on this review, we decided to adopt and implement a Progressive Web App (PWA) strategy as an alternative.

The decision was driven by several practical factors. A PWA reduces duplicated development effort by allowing the team to maintain a single codebase instead of continuing to build and support separate web and mobile implementations. It also simplifies deployment and updates, since improvements made to the web platform become immediately available across devices without requiring separate mobile release cycles. In addition, a PWA helps preserve a consistent user interface and workflow across desktop and mobile usage, which is particularly important for scheduling and task-based operational tools.

Following this decision, the foundational PWA features were implemented in the platform, including the application manifest, installable configuration, and service worker setup. This allows the system to be installed on mobile devices and accessed directly from the home screen, enabling a more app-like experience while continuing to use the same web-based architecture.

Role-Based Actions Implementation

During this period, role-based actions were implemented to ensure that different types of users interact with the Eco Clean platform according to their responsibilities. The system now distinguishes between administrative users and staff members, allowing each role to access only the features and actions relevant to their tasks. Administrative users can manage core entities such as users, clients, staff assignments, and scheduling configurations, while staff users are primarily focused on viewing assigned tasks and interacting with operational scheduling features.

This role-based structure improves both system security and workflow clarity by preventing unauthorized actions and reducing interface complexity for different user types. By enforcing permissions at the application level, the platform ensures that sensitive operations such as managing users or modifying system data are restricted to authorized roles. This implementation helps align the software with real business operations while creating a more structured and maintainable access control model for future feature expansion.

System Stability and Deployment Improvements

Additional work focused on stabilizing the backend infrastructure and preparing the system for production environments. This included resolving database schema inconsistencies, improving API logic, and strengthening the

interaction between frontend components and backend services. These improvements helped ensure that scheduling workflows and entity management operate reliably under real usage conditions.

Overall Progress

Overall, the work completed this week further improved the maturity and functionality of the Eco Clean platform. The application now includes a fully functioning calendar scheduling interface, structured management of users, clients, and staff, and the initial implementation of a Progressive Web App (PWA) to support mobile accessibility. The PWA setup allows the system to be installed and used on mobile devices, bringing the platform closer to a unified web-and-mobile experience.

In addition, role-based actions and access control were implemented to ensure that different types of users (such as administrators and staff members) interact with the system according to their responsibilities. This helps maintain proper operational workflows and improves system security. Several stabilization improvements were also made across the application, including refinements to navigation, data handling, and UI consistency to ensure the system behaves reliably under real usage conditions.

Together, these developments move the project closer to a production-ready operational tool while establishing a more scalable architecture for future expansion.

Work Logs

Date	Time	Description
2026-03-05	10:00-11:30am	Implemented initial PWA configuration including manifest setup and installable configuration
2026-03-05	11:30am-1:00pm	Generated required PWA icon assets and configured maskable icons for mobile installation
2026-03-05	2:00-3:30pm	Debugged manifest loading issues and corrected manifest placement for browser recognition
2026-03-05	3:30-5:00pm	Implemented role-based routing logic to redirect admin and staff users to appropriate dashboards
2026-03-05	6:30-8:00pm	Created staff task route structure and began development of mobile task interface for PWA
2026-03-06	9:00-10:30am	Designed bottom navigation bar pattern optimized for mobile PWA usage

2026-03-06	10:30am-12:00pm	Tested PWA installability and mobile behavior using local network device testing
2026-03-06	2:00-3:30pm	Stabilized navigation behavior for role-based access control across admin and staff routes
2026-03-06	4:00-5:30pm	Improved UI responsiveness and layout adjustments for mobile PWA screens
2026-03-07	9:30-11:00am	Implemented initial staff task interface structure and scheduling overview for PWA workflow
2026-03-07	11:00am-12:30pm	Refined PWA configuration and verified manifest, icons, and service worker integration
2026-03-07	7:30-10:30pm	Extended job notes system and improved task documentation workflow
Total	19h 30m	

AI Usage

AI	Prompt	Value Addition
ChatGPT 5.3, Plus	Implementing Progressive Web App (PWA) in Next.js application	Enabled installable web application with manifest configuration and service worker support
ChatGPT 5.3, Plus	Generating PWA icon sizes (192x192, 512x512, maskable icons) from base logo	Created compliant icon assets required for PWA installation and mobile compatibility
ChatGPT 5.3, Plus	Debugging manifest.webmanifest not appearing in browser devtools	Resolved manifest placement and configuration issues for proper PWA recognition
ChatGPT 5.3, Plus	Implementing role-based routing for admin vs staff users in Next.js	Enabled conditional navigation and protected routes based on user role
ChatGPT 5.3, Plus	Redirecting authenticated users to role-specific root pages	Created structured routing logic for admin dashboard vs staff task interface
ChatGPT 5.3, Plus	Building mobile-friendly task interface for PWA staff workflow	Designed simplified navigation structure suitable for mobile task execution
ChatGPT 5.3, Plus	Designing bottom navigation bar for PWA interface	Improved usability for mobile staff users with thumb-access navigation

ChatGPT 5.3, Plus	Testing PWA functionality on local network mobile devices	Enabled real-device testing through local network hosting configuration
ChatGPT 5.3, Plus	Fixing timezone inconsistencies between database and frontend display	Ensured accurate appointment scheduling using consistent UTC timestamp handling

Repo Check-in

Work on the repository was carried out on a regular basis, aligned with the project timeline and development milestones. Development activities included frontend UI components, backend API implementation, database schema evolution, PWA components and deployments.

Majority of the work for the eco-clean-web was committed through the branch **pwa and extend-notes (work in progress)**.

PR #9: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/9 - *Technical details of this milestone is included in this PR*

The work was checked into the repository almost every day, followed by a merge to main almost every 7 days to publish changes to the live preview.

